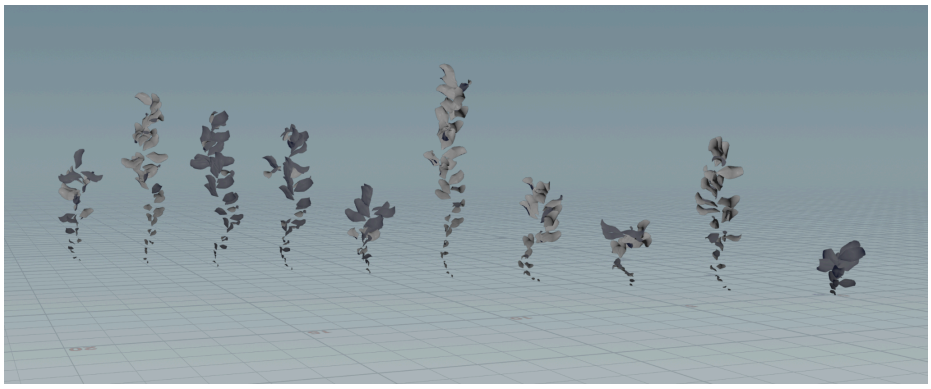
Summary Of Quarter
Vellum Explorations for Plant Growth

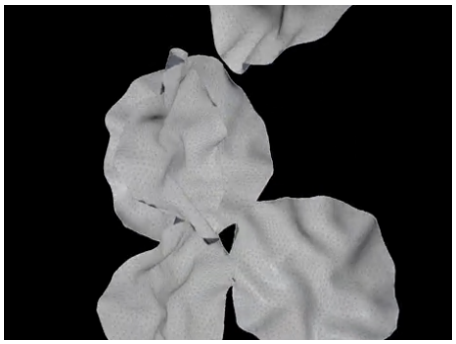
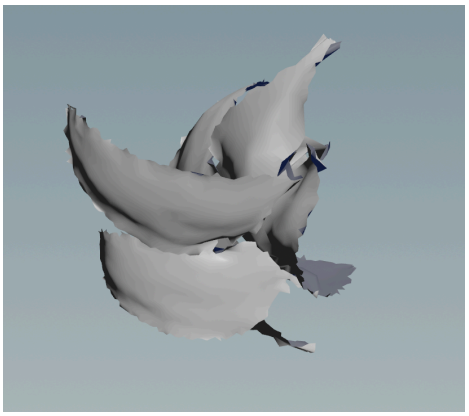
1. Vellum Successes
    a. Using multiple “petals” to force internal “petals” together, and then blasting so that only two leaves remain
    b. Timeshift in a for each loop containing a copy to points offsets the animation for every leaf bud- This will happen before the vellum simulation has run
    c. Hand animating stem was done in addition to the simulation to get more control of the movement
        i. Combination of Layered simulations and hand animating has gotten me to the closest result this quarter
        ii. Having a basic Rig on the plant was crucial because the vellum is ultimately following the animation
    d. Without concerning about growing from the bud, PDG worked very well with Vellum Vine tests to create variations of vines with different curves and number of leaves



    e. Vellum rest blend was crucial to having the simulation follow the hand animation- as long as starting frame has no intersections, the rest followed
    f. Pinning minimal points on the leaves, just enough for the vellum to follow the animation, but little enough that there is minimal intersection between the Pinned points
        i. The balance between pinning enough points, but not too much so that there is natural movement, hard pins would allow the leaves to stick to the stem better, but soft pins allowed for more natural movement
    g. Dampness is an important aspect of Vellum- controls bounciness, balance of stiffness and bend is most important
    h. Point Deform to deform detailed geometry to simulation OR for a simple extrusion Vellum Post Process will also work

2. Limitations of Vellum
    a. Starting frame has to have some geometry present in order for the vellum sim to work
        i. Possible Solution
            1. Set Simulation Start Frame to after geometry shows up
            2. Delete based on Curveu after the simulation is complete
    b. Deforming shape because of scale increase

i. Starting from open position and going towards the closed position gave better results than starting small and growing
ii. Balance of mesh density based on scale was hard to find without the vellum distorting at some point

c. Trying to use Feather tool to act as leaves since it is built to not intersect
   i. Once the Feather was meshed it would stop behaving like a feather- looking to play around with inside the feather tools to see if some tweaks can edit that

3. Stem
   a. Branching stems need to be a solid piece of geometry for subsurface to not cause any artifacting
      i. Possible Solutions
         1. Converting ONLY Stems to VDB
            a. Issues: Hard to apply UVS, point count changes per frame → therefore difficult to add Hair OR bring into Unreal as a VAT
         2. Create branching topology for polyWire
            a. Issues: Point count still changes per frame

b. How can the stem growth seem more fluid?
    i. Possible Solution
        1. Stem is a Vellum Soft Body sim
c. Branching stems to grow by position
    i. The Carve tool would grow all the curves at the same time
    ii. Solution: VEX solution to sort point numbers based on position, and then delete points based on that

4. Combining Vellum & Gaussian Splats
    a. Biggest limitation was that Gsplats do not have an alpha which would be the basis of a lot of compositing
        i. Lot of Roto was done to get around this

5. Questions left to answer
    a. Is there a more elegant solution to creating the connection between the stem and the leaves
    b. For a close up shot, would we get the best results by hand animating?
    c. Is it possible to remesh geometry as the simulation is running?
    d. For real time, how to optimize without reducing the quality?